FPGA Hackathon 2026

With the rapid growth of intelligent systems at the network edge, there is an increasing need for low-latency, low-power, and real-time AI inference directly on embedded hardware. Cloud-based AI solutions suffer from bandwidth limitations, high latency, privacy concerns, and increased energy consumption.

FPGA Hackathon 2026 focuses on Edge AI implemented on AMD/Xilinx FPGA platforms, encouraging participants to explore hardware-accelerated Edge AI inference engines by developing innovative, efficient, and application-driven solutions, implemented at the Verilog RTL level. The event emphasizes research-driven innovation and the development of industry-relevant skills, providing a competitive platform for undergraduate, postgraduate, and doctoral students across India.

Problem Statement

Participants are required to design and implement hardware-accelerated Edge AI solutions on AMD/Xilinx FPGA platforms using Verilog RTL, targeting real-world applications such as Agriculture, Biomedical Systems, Traffic Management, Smart Energy Systems, and Defense. Any other suitable real world application domain can also be considered. The solutions should emphasize real-time performance, energy efficiency, reliability, and on-device intelligence.

Hackathon at a Glance:

- **Eligibility:** BE, ME, MSc, and PhD students (open to students from all institutions across India)
- **Team Size:** 2 members per team
- **Prize Pool:** Up to ₹1.5 Lakhs
- **Entry fee:** Nil
- The hackathon will be conducted in two rounds.
- **Round 1 (online):** focused on hardware-accelerated Edge AI solutions using Verilog RTL
- **Round 2 (offline):** 6–8 teams shortlisted for the On-site Hackathon, and 10–15 teams shortlisted for the Poster Presentation during the final event.
- All participants qualifying for Round 2 will receive certificates
- Opportunity for industry interaction and national-level visibility

- **Registration Form:** https://forms.gle/X43LWxbF3oZSpFm78

Round 1: Online Screening

- Develop an AI/ML model for a chosen real-world Edge AI application using suitable software tools (Python, MATLAB, C/C++, etc.) for training and validation. Use standard and publicly available datasets.
- Implement the inference stage of the model in RTL (Verilog) as a hardware accelerator.
- Demonstrate how the AI model operates at the edge emphasizing on real-time performance, low power consumption, and efficient hardware utilization.
- FPGA prototyping using AMD/Xilinx FPGA platforms is strongly encouraged and will be given preference; however, correct RTL designs with validated simulation results will also be considered.
- Shortlisted teams will advance to Round 2 for an on-site Hackathon or Poster presentation.

- **Round 1 Submission Link:** https://forms.gle/utP39b3sj4c3B4eGA

Round 2: Offline Final Competition (2 Days)

- **Venue:** *BITS Pilani, Hyderabad Campus*

Day 1 – On-site Hackathon

- Finalist teams will participate in the grand innovation challenge and deploy the inference stage of edge AI model for the chosen application at RTL level using real-time data and/or sensor interfacing on AMD/Xilinx FPGA platform with emphasis on real-time performance, low power consumption, and efficient hardware utilization.
- AMD/Xilinx FPGA boards, including ZedBoard, Zybo, and PYNQ, will be available for Round 2. Sensor and wireless interface modules (PMODs) will also be provided during the offline event.
- Teams may be required to implement an extended or modified problem statement during the hackathon.
- Industry/academia expert talks and hands-on workshop on hardware-accelerated Edge AI will be held.

Day 2 – Presentations & Outreach

- Technical presentations by hackathon finalist teams, poster presentation session by shortlisted poster teams, industry/academia expert talks, panel discussion, Live-Demo/ hands-on workshop, final evaluation and announcement of results.

Application Domains:

1. **Agriculture:** Modern agriculture increasingly relies on real-time data to improve yield, reduce waste, and optimize resource usage. Edge AI enables intelligent decision-making directly at farms, even in remote areas with limited connectivity. Proposed solutions may include, but are not limited to, crop disease detection from leaf images or sensor data, soil moisture classification for automated irrigation, and crop yield prediction using environmental parameters.
2. **Biomedical Systems:** Healthcare systems rely on wearable and portable medical devices that require local, real-time data processing to ensure patient privacy and rapid response. Proposed solutions may include, but are not limited to, ECG/EEG signal classification for abnormality detection, heart rate anomaly detection, and medical image feature classification.
3. **Traffic Management System:** Urban traffic congestion leads to increased fuel consumption, pollution, and delays. Edge AI enables localized, real-time traffic decision-making without reliance on centralized servers. Proposed solutions may include, but are not limited to, vehicle density estimation at intersections, adaptive traffic light control, and accident or congestion detection.
4. **Smart Energy System:** Energy efficiency is a critical concern in smart buildings, factories, and smart grids. Edge AI enables intelligent, on-device energy management by optimizing resource usage without constant reliance on cloud connectivity. Proposed solutions may include, but are not limited to, load prediction and peak demand detection, as well as occupancy-based lighting or HVAC control.
5. **Defense System:** Defense applications demand real-time, low-latency, and secure data processing in resource-constrained, mission-critical environments. Edge AI enables on-device intelligence that ensures fast response, reliability, and data security without reliance on cloud connectivity. Proposed solutions may include, but are not limited to, target detection or classification using sensor or image data, anomaly detection in radar, sonar, or surveillance signals, and multi-sensor data fusion for autonomous threat detection.

Final submission and Evaluation:

- Complete Verilog RTL source code (project files in ZIP format)
- Video demonstration of implemented work (5–10 minutes).
- Technical report covering application overview and AI model architecture, design methodology, RTL Architecture, novelty and advantages over existing solutions. Please adhere to the format given below.
  Round 1 report format
- Evaluation will be based on relevance to real-world problems, technical complexity, novelty, quality of hardware implementation, performance/metric comparison, completeness, and clarity of presentation.

Important Dates:

- Round 1 Problem Statement Release: 4 February 2026
- Registration Deadline: 1 March 2026
- Round 1 Submission Deadline: **29 March 2026**
- Round 1 Result Declaration: TBD
- Round 2 (Offline Event): 10–11 April 2026

Sponsorship & Organization:

- **Sponsored by:** AMD
- **Co-sponsored by:** Institution's Innovation Council (IIC), BITS Pilani Hyderabad Campus
- **Hosted by:** BITS Pilani
- **Organized by:** Department of EEE and IEEE Student Branch, BITS Pilani Hyderabad Campus

Organizing Committee:

- **Faculty Coordinator:** Prof. Amit Kumar Panda, Dept. of EEE, BITS Pilani Hyderabad Campus
- **AMD Coordinator:** Mr. Santosh Yachareni, Senior Director, Analog Division, AMD
- **Student Coordinator:** Sree Harsha, IEEE
- **Student Co-Coordinator:** Sharundhathya, IEEE

Timeline

Registration Deadline
20 Feb 2026

Round 1 submission
15 March 2026

Round 1 result
20 March 2026

Round 2 Hackathon
10 - 11 April 2026

Sponsorship & Organization

Faculty: Prof. Amit Kumar Panda
AMD: Mr. Santosh Yachareni
Student Coordinator: Sharundhathya C, IEEE Student Branch
Student Co-Coordinator: Sree Harsha G, IEEE Student Branch

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Co-Sponsored by ***IIC, BITS Pilani Hyderabad***

Key Focus Areas

AMD/Xilinx FPGA
Leverage AMD/Xilinx architectures to achieve maximum throughput and energy efficiency for real-time intelligence.

Real-time performance
Achieve maximum processing speed and low latency for real-time applications.

Efficient hardware utilization
Maximize the use of FPGA resources to create compact and powerful solutions.

FPGA Hackathon 2026

₹ Up to 1.5 Lakh Prize Pool

- Progress from an Online Screening to an Offline Grand Finale at BITS Pilani, Hyderabad Campus
- Compete for a **₹1.5 Lakhs prize pool** with **no entry fee**
- Network with experts and gain national-level visibility.

About the **Hackathon**

Round 1

Participating teams must choose a real-world application and develop a robust AI/ML model using software tools like Python, MATLAB, or C/C++. This stage focuses on implementing the inference engine in Verilog RTL as a dedicated hardware accelerator, validated through simulation or FPGA prototyping. Submissions will be evaluated on technical complexity, novelty, and the ability to demonstrate efficient hardware utilization for real-time edge performance.

Round 2

Day 1 – On-site Hackathon
Finalist teams will take on the grand innovation challenge, deploying the inference stage of their Edge AI models at the RTL level using real-time data and sensor interfacing on AMD/Xilinx FPGA platforms. Includes bonus sessions with expert talks and a hands-on workshop on hardware-accelerated Edge AI.

Day 2 – Presentations & Outreach
Features technical presentations, poster sessions, expert talks, live demos, and the final results announcement.

Participation

Team Size: 2 members per team
Eligibility: BE, ME, MSc, and PhD students from all institutions across India
Zero Entry Fee
Travel reimbursement: Coverage of up to **₹3,000 per student** for both team members selected for Round 2
Poster presentation: Submissions for round 1 will also be considered for a poster presentation

Up to 1.5 Lakh Prize Pool

Application **Domains**

Traffic Management System:

Edge AI facilitates localized, real-time decision-making to reduce congestion and pollution

Smart Energy System

Intelligent, on-device resource optimization for buildings and grids without relying on the cloud

Agriculture

Edge AI for smart farming — crop disease detection, soil moisture classification, and yield prediction.

Biomedical Systems

Real-time health monitoring — ECG/EEG anomaly detection, heart rate analysis, and medical image classification.

Defense Systems

Mission-critical intelligence — target detection, anomaly detection in radar/sonar, and multi-sensor data fusion.

Submission & Evaluation

Complete Verilog RTL source code
5-10 minute video demo
Technical report covering the application, AI model, design, and novelty.

Contact & Venue

BITS Pilani, Hyderabad Campus

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